MSCS2202 Machine Learning Kaggle Challenge

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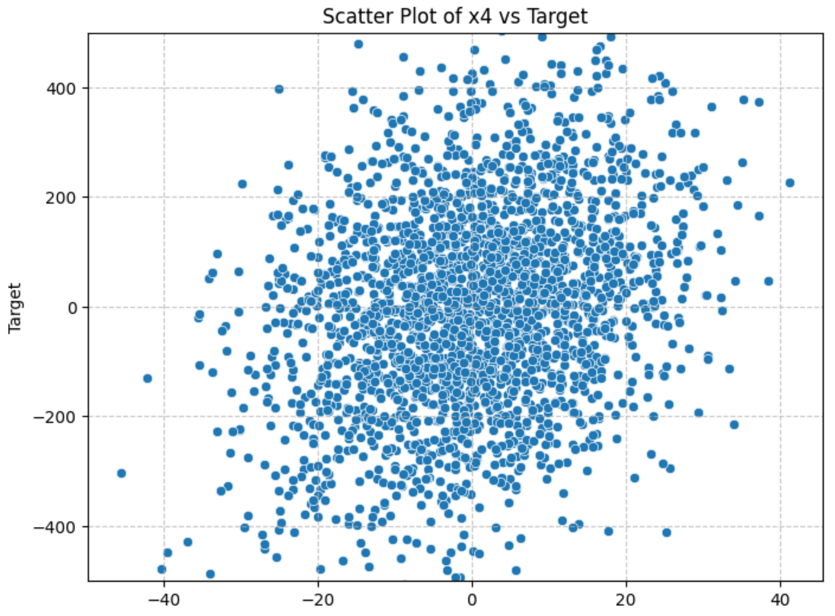
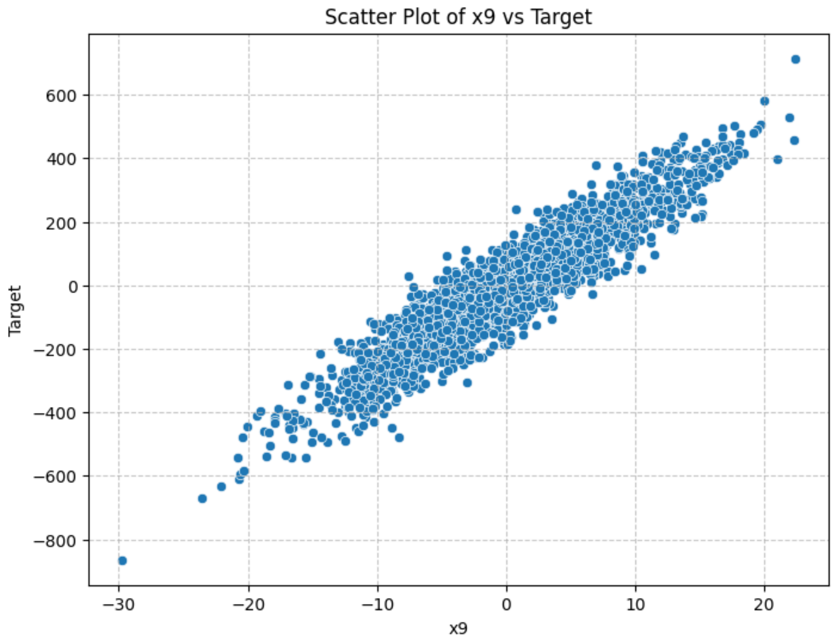
Main Ideas & Results

- SVR with sigmoid kernel generalizes best on the noisy, outlier-heavy dataset.

- **Final Model Selected:**
  - SVR (Support Vector Regression)
    - Kernel = sigmoid
    - C = 2000.00
    - ε = 379.274
    - γ = auto
- **Pipeline:**
  - Imputer = Mean
  - Scaler = Robust Scaler
  - 3-fold CV
- **Results:**
  - Public Test R^2 Scoring: **0.04751**
  - Private Test R^2 Scoring: **0.04693**



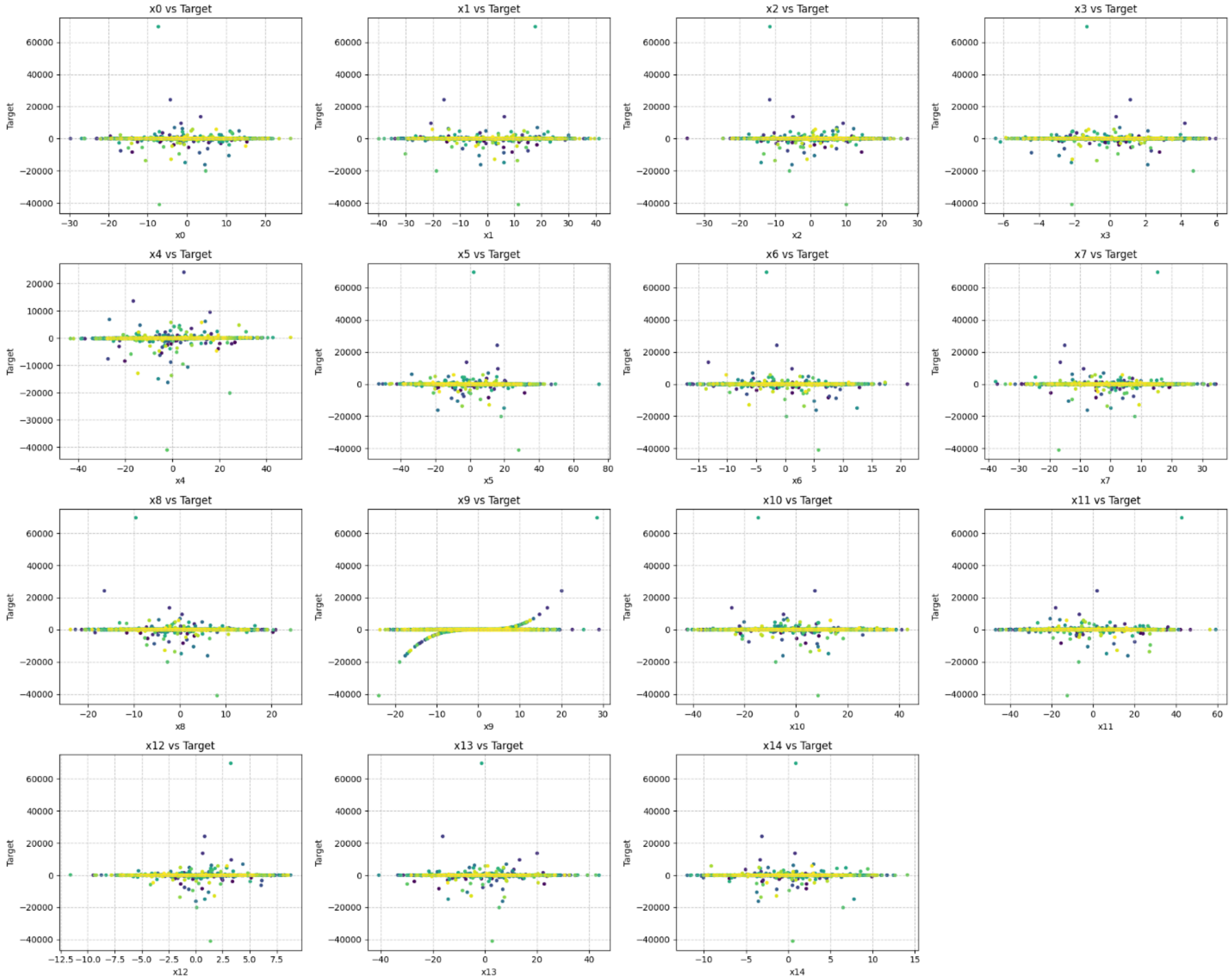


Training Data Exploration

	#NA	MEAN	STD	MIN	25%	50%	75%	MAX
x0	32	0.06	7.85	-29.71	-4.96	-0.03	5.41	26.42
x1	30	0.12	12.49	-39.97	-8.57	0.06	8.49	41.26
x2	22	0.05	7.85	-34.75	-5.26	0.04	5.37	27.21
x3	24	-0.04	2.02	-6.43	-1.38	-0.05	1.32	5.94
x4	34	0.11	12.79	-43.40	-8.60	-0.47	8.49	50.20
x5	28	-0.46	15.18	-52.68	-10.53	-0.96	9.51	74.72
x6	30	0.04	5.42	-16.94	-3.60	0.01	3.63	21.15
x7	30	0.17	10.26	-37.57	-6.67	0.19	7.17	34.32
x8	34	0.14	6.91	-23.79	-4.47	0.13	4.88	24.13
x9	30	-0.22	7.00	-23.89	-4.92	-0.25	4.56	28.98
x10	31	-0.25	13.75	-42.32	-9.34	-0.12	8.80	43.26
x11	32	-0.35	14.47	-46.87	-9.95	-0.59	9.21	59.11
x12	38	-0.07	2.96	-11.60	-2.09	-0.04	1.92	8.79
x13	28	0.16	11.27	-40.71	-7.33	0.20	7.42	43.76
x14	25	0.02	3.87	-11.97	-2.57	0.07	2.49	14.10
target		-22.04	1978.18	-41008.01	-43.17	-0.77	42.00	69628.20

Summary

- 15 Numerical features with ~1% missing values each
- Feature scales vary (STD 2-15)
- Target value
  - Small Mean value
  - Large STD
  - Range spans 110k+, dominated by extreme outliers
  - Middle 50% of target values lie within ±50

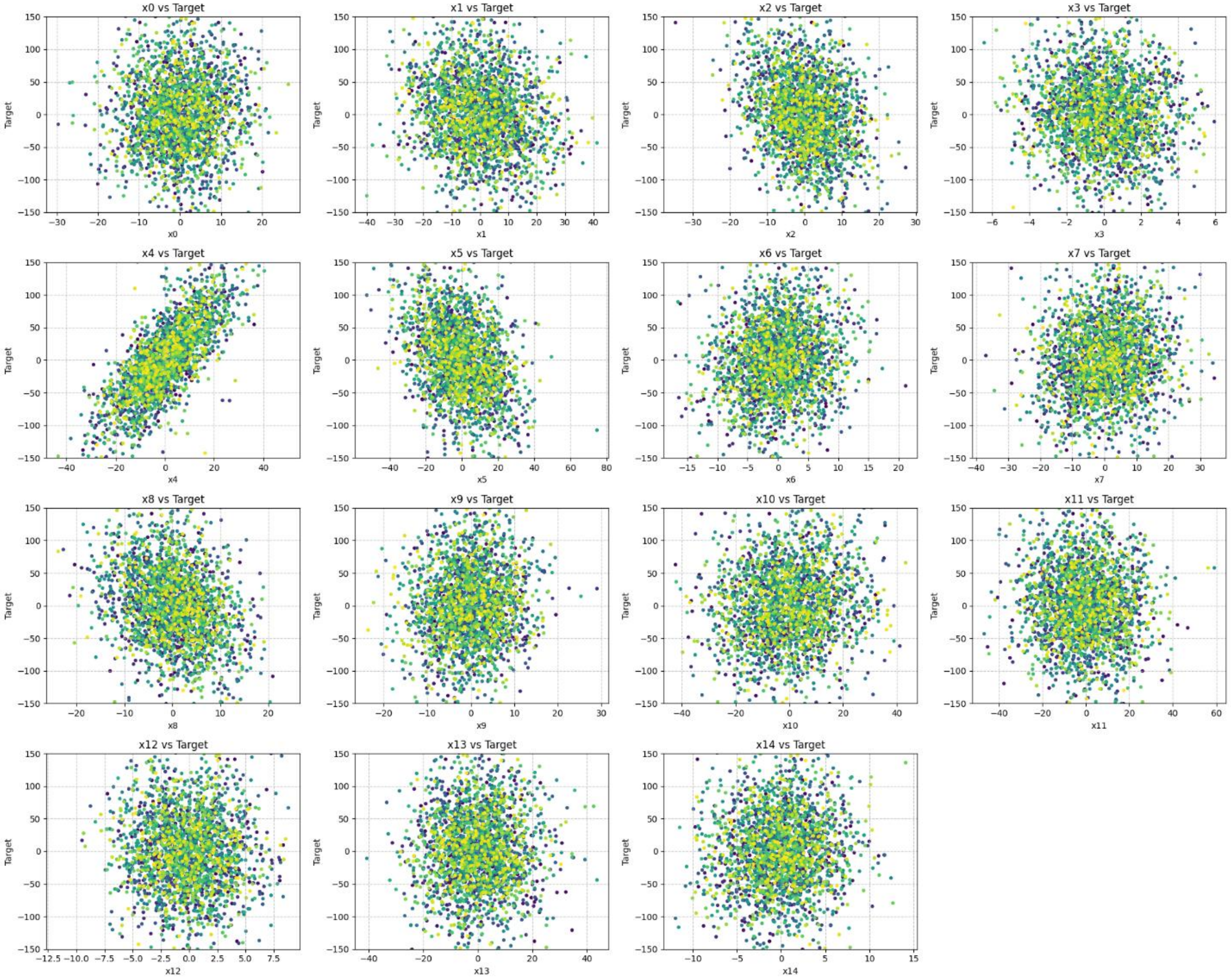


x0-x14 vs. target (full range)

- Over 90% of the target value centered within +-200
- Extreme values are huge, may be dominating noise
- x9 showing some polynomial relationship between target and x9
- Other features show no clear relationship with target

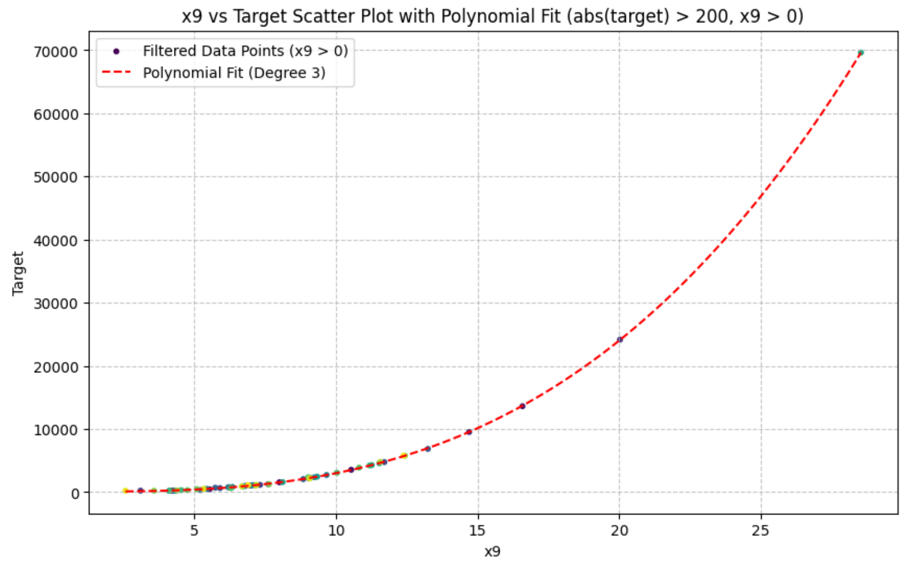
x0-x14 vs. target (zoom to +-150)

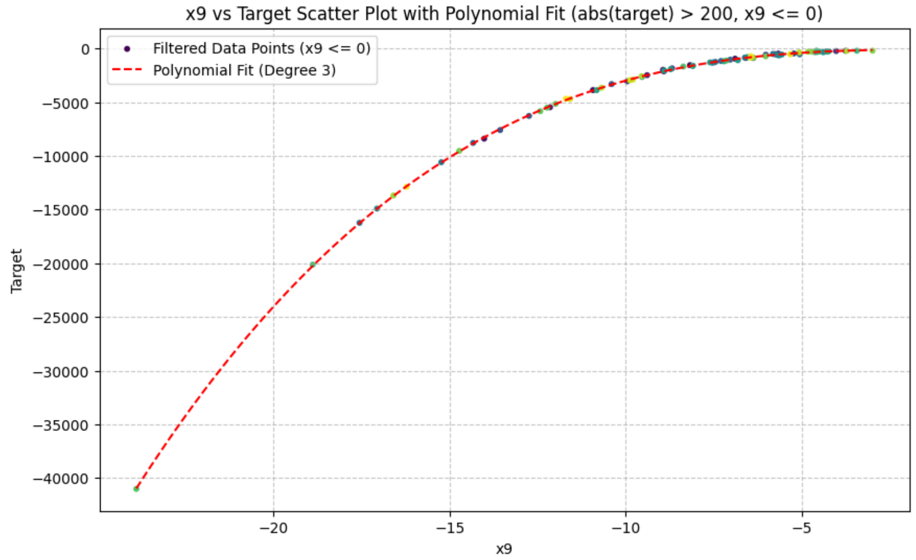
- Reveals subtle linear trends in x4, x5, x8
- Most features remain noise-like



EDA Summary

- Suggests non-linear modeling and robust preprocessing





- Missing values are moderate → mean/median imputation
- Feature scales vary → RobustScaler preferred
- Target has extreme outliers → linear models unstable
- x9 shows degree of 3 polynomial-like structure
- x4, x5, x8 show weak linear trends (visually)
- Most features do not show clear relationship

Initial Standard Modeling Attempts

- Imputer: Median
- Scaler: Standard/Robust Scaler
- 10-fold Cross-Validation

Model	Public Test R^2
Linear Regression (OLS)	-0.09273
Random Forest	-0.11347
SVR (poly kernel)	0.00550
XGBoost	0.00442
Elastic Net SVR	0.00111

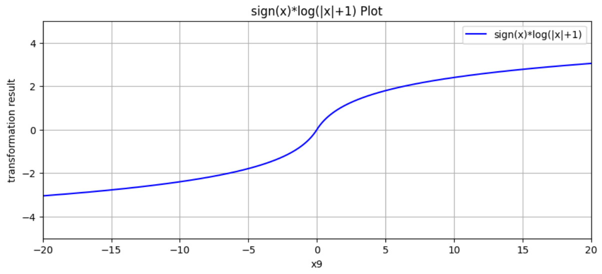
Takeaway: These models failed to capture nonlinear, heavy-tailed structure

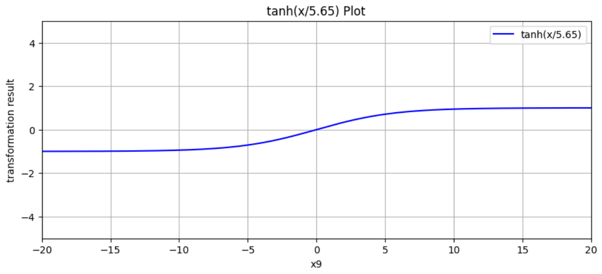
Non-liear Transforms for x9

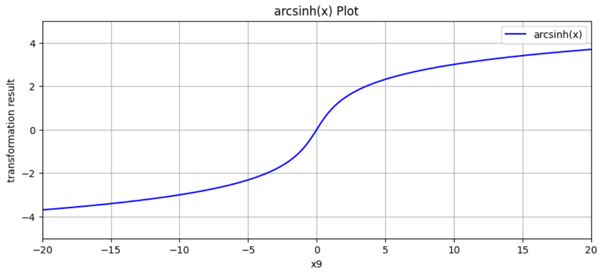
Transformers

- $x9^2$, $x9^3$
- sign(x9)* $x9^2$, $x9^3$
- sign(x9)*log(abs(x9)+1)
- tanh(x9/5.65)
- arcsinh(x9)
- SplineTransformer

Huber Regression Model





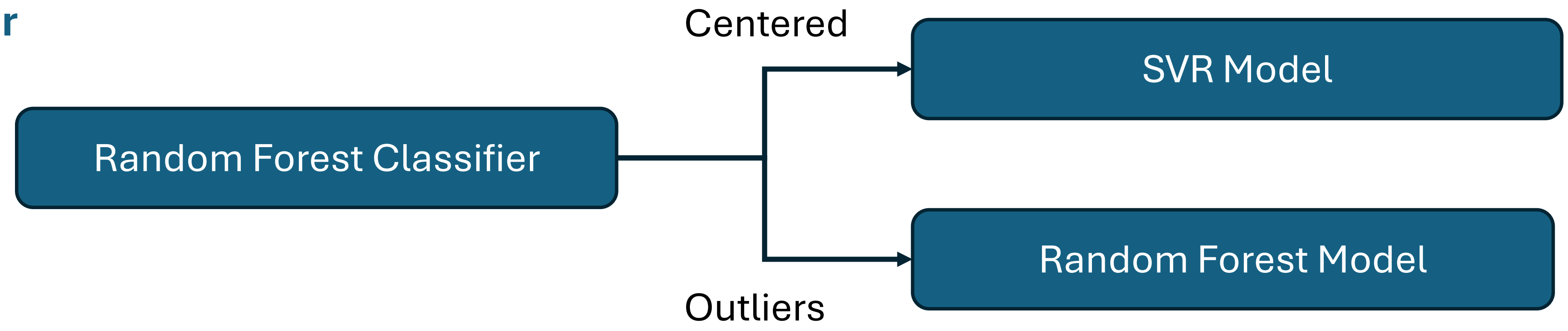


Observation: Training score improved (0.06-0.13), but public test R^2 remained ≈ -0.05-0.03.

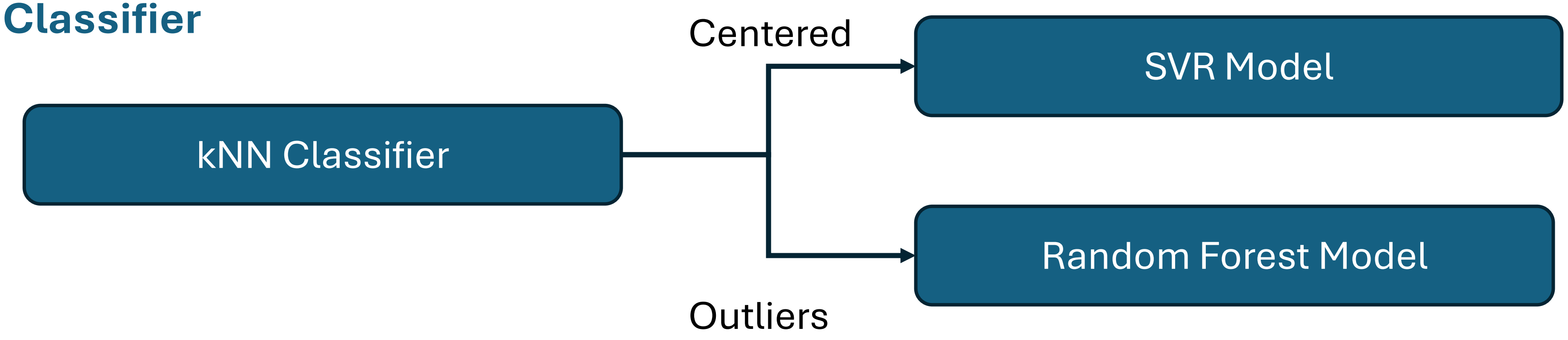
Thoughts: Higher score on train set but low/negative scores on public test may suggest an overfit. Non-linear transformations could potentially improved training but not test performance.

Two-Stage Model: Classifier + Regression

RF Classifier



1-NN/2-NN Classifier



Observations: Classifier and Regressors each can achieve high R^2 score (>0.8), but bad performance on public test set (~0.005).

Conclusion: The classifier is very likely overfit, and this approach did not pick up meaningful pattern.

OLS Linear + SVR Ensemble

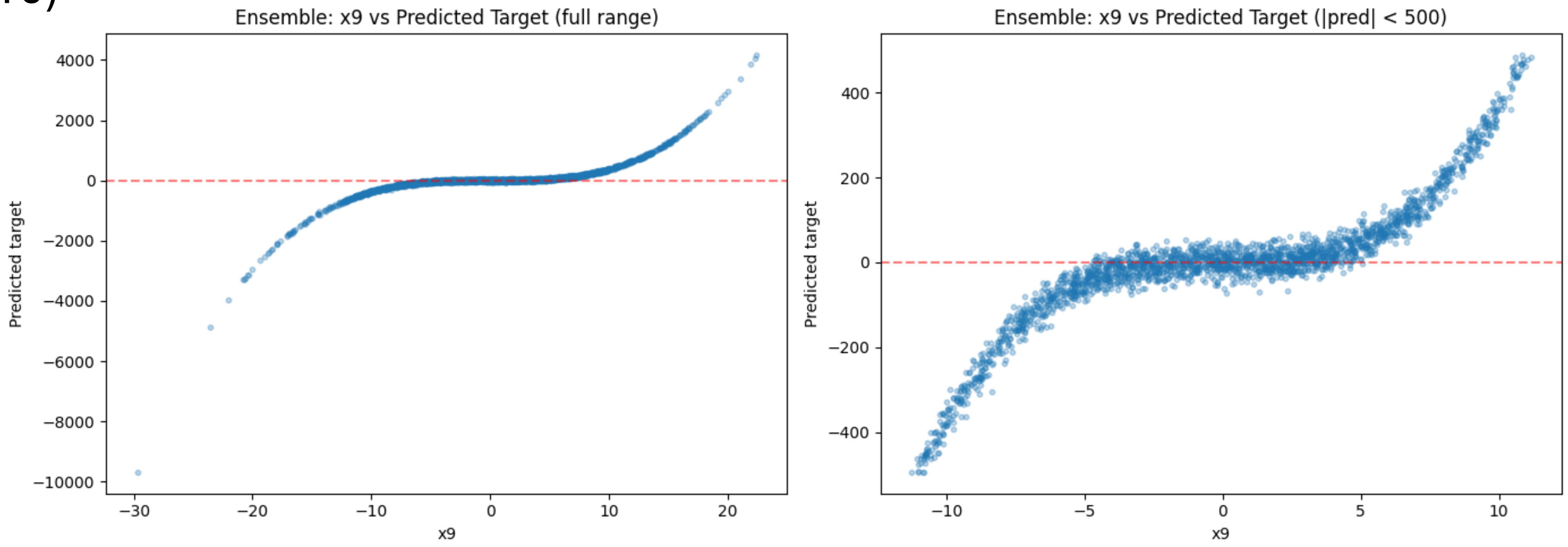
- Imputer: Median
- Scaler: Standard/Robust Scaler
- 10-fold Cross-Validation
- OLS Model with $x9^3$ for poly features
- SVR - rbf kernel
- Weight sweep (10/90 → 90/10)

Results

Final Selected: 60% OLS + 40% SVR

Best CV R^2 ≈ 0.1077

Public test R^2 = −0.1931

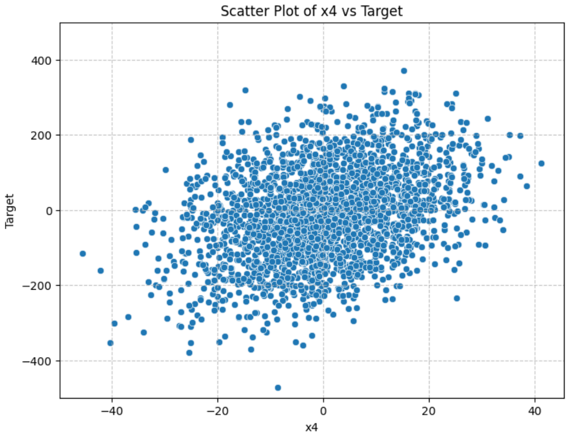


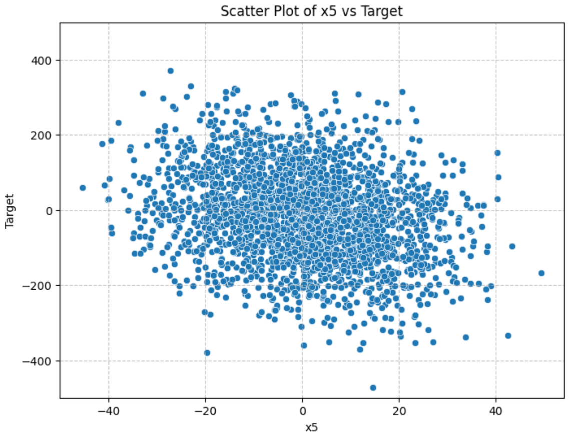
Hyperparameter Search for SVR Optimization

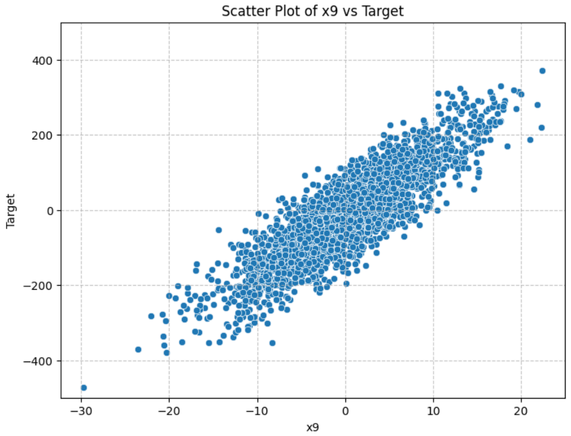
- Imputer ∈ {mean, median}
- Scaler ∈ {none, Standard, Robust}
- Kernel ∈ {rbf, sigmoid}
- CV: 10 -folds
- γ ∈ {auto, scale}
- Coarse exponential search for C, ε

Best Model from Fine Search

- Imputer = Mean
- Scaler = Robust
- Kernel = rbf
- γ = auto
- C = 356, ε = 213







Results

Best CV R^2 = 0.0325

Public test R^2 = 0.0406

Hyperparameter Search and Reduce CV folds

- Outlying extreme target < 200 data points → 10-fold may introduce unbalanced splitting by chance that may impact CV scoring
- Reduce CV folds from 5 to 4, 3, 2 for stability improvement

	CV R^2	Refit R^2	Public Test R^2
4-folds	0.05027	0.05095	-0.0799
3-folds	0.04206	0.04269	**0.04751**
2-folds	NA	0.05203	0.02623

← Final Model Selected

Takeaway: Fewer folds reduce the chance of uneven outlier distribution across splits.

Results Summary and Model Comparison

Model	Train set R^2	Public Test R^2	Private Test R^2
Random Forest	0.01625	-0.11347	0.17518
Linear OLS	0.02660	-0.09273	0.06244
RF Classifier + SVR/RF Model	-0.01256	0.00496	0.01084
Spline Trans + Huber	0.1100	0.01139	0.11478
OLS + SVR Ensemble	0.1077	-0.19307	0.16230
10-fold SVR	0.03258	0.04064	0.02927
3-fold SVR [Final Selection]	**0.04269**	**0.04751**	**0.04693**
2-fold SVR	0.05203	0.02623	0.06055

Most of the linear model shows instability according to the large discrepancy between train and test R^2.

I selected the **3-fold SVR model** with sigmoid kernel for the most stable and generalized performance across linear and non-linear features for the dataset.

Thank You!
Q&A

GitHub link: Github-->ml_kaggle_challenge_2026